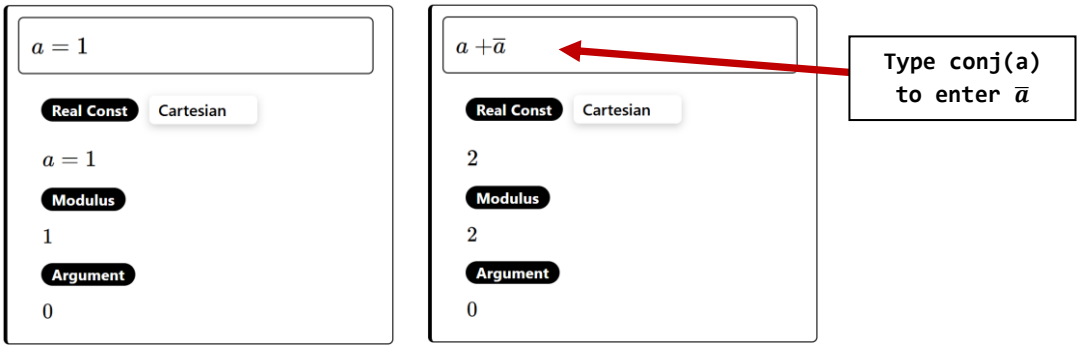
Conjugates Geometry

- Open **Komplexiti** (https://www.korovatron.co.uk/komplexiti/)
- Add the **constant** $a = 1$ to the panel.
- Add the **expression** $a + \overline{a}$ to a second card.



For each of the following expressions use the mouse to **drag** the point **a** around the Argand diagram and observe what happens.

$a + \overline{a}$	$a - \overline{a}$
Describe how $a + \overline{a}$ behaves as you drag the point a around the Argand diagram. It is confined to the real axis, twice as far from the imaginary axis as a. **Can you prove what you have discovered?** $a = x + yi$ $\overline{a} = x - yi$ $a + \overline{a} = x + yi + x - yi$ $a + \overline{a} = 2x$	**Describe how $a - \overline{a}$ behaves as you drag the point a around the Argand diagram.** It is confined to the imaginary axis, twice as far from the real axis as a. **Can you prove what you have discovered?** $a = x + yi$ $\overline{a} = x - yi$ $a - \overline{a} = x + yi - (x - iy)$ $a - \overline{a} = 2yi$

Conjugates Geometry

$a\overline{a}$

Describe how $a\overline{a}$ behaves as you drag the point a around the Argand diagram.

It is confined to the positive real axis.

Can you prove what you have discovered?

$$a = x + yi$$
$$\overline{a} = x - yi$$

$$a\overline{a} = (x + yi)(x - yi)$$

$$a\overline{a} = x^2 + y^2$$

$$\therefore a\overline{a} \geq 0$$

$\frac{a}{\overline{a}}$

Describe how $\frac{a}{\overline{a}}$ behaves as you drag the point a around the Argand diagram.

It is confined to the circle centre (0, 0), radius 1.

Can you prove what you have discovered?

$$a = x + yi$$
$$\overline{a} = x - yi$$

$$|a| = \sqrt{x^2 + y^2}$$
$$|\overline{a}| = \sqrt{x^2 + y^2}$$

$$\left|\frac{a}{\overline{a}}\right| = \frac{|a|}{|\overline{a}|} = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2}} = 1$$

$\therefore \frac{a}{\overline{a}}$ is equidistant from the origin,

so is a circle centre (0, 0), radius 1.